

A READER’S GUIDE TO “THE EQUIVARIANT COHOMOLOGICAL HIKITA CONJECTURE FOR ARBITRARY QUIVERS”

NOTES PREPARED FOR THE SLMATH QFT PROGRAM

Caveat. These notes summarize a preprint released by OpenAI on 24 September 2026 [1] as part of a collection of manuscripts written by an unreleased internal model. The preprint has not been refereed and has no Lean formalization in the release; the release notes themselves say that unformalized results “could have issues.” These notes were drafted by Claude (Anthropic) at Ben Webster’s request from the L^AT_EX source, and reorganize the argument for readability. They do *not* verify the proofs: every “proof” below is a sketch of what the preprint claims, and the words “the paper shows” should be read as “the paper asserts, with an argument I have summarized but not checked.” Section 8 records which inputs are standard and which steps are new.

1. THE STATEMENT

1.1. Data. Fix a finite quiver $Q = (Q_0, Q_1)$, loops and multiple arrows allowed, with gauge dimension vector $\mathbf{v}$ and framing vector $\mathbf{w}$. Put

$$G = \prod_{i \in Q_0} \mathrm{GL}(V_i), \quad N = \bigoplus_{a: i \rightarrow j} \mathrm{Hom}(V_i, V_j) \oplus \bigoplus_i \mathrm{Hom}(V_i, W_i),$$

with $\dim V_i = v_i$, $\dim W_i = w_i$. Let F be any torus acting on N and commuting with G (a “flavor torus”), and let $\mu : T^*N \rightarrow \mathfrak{g}^*$ be the moment map.

Fix a character ν of G . The paper calls ν *regular* if, on $\mu^{-1}(0)$, every ν -semistable point is stable and G acts freely on the semistable locus. Then

$$X = \mu^{-1}(0)^{\nu\text{-st}}/G$$

is a smooth Nakajima quiver variety, possibly empty.

1.2. The two sides. *Higgs side.* $H_F^*(X; \mathbb{C})$, the F -equivariant cohomology.

Coulomb side. Let $\mathcal{O} = \mathbb{C}[[t]]$, $\mathcal{K} = \mathbb{C}((t))$, and let $\mathcal{R} = \{[g, s] \in G(\mathcal{K}) \times_{G(\mathcal{O})} N(\mathcal{O}) : gs \in N(\mathcal{O})\}$ be the Braverman–Finkelberg–Nakajima space of triples. Its $G(\mathcal{O}) \rtimes F$ -equivariant Borel–Moore homology

$$A_F = H_*^{G(\mathcal{O}) \rtimes F}(\mathcal{R})$$

is a commutative algebra under convolution (there is no loop-rotation parameter, so this is the *commutative* Coulomb branch, deformed by the flavor parameters). Its spectrum $Y_{\mathfrak{f}} = \mathrm{Spec} A_F$ is the flavor-deformed Coulomb branch. The algebra A_F is graded by $\pi_1(G) = \mathbb{Z}^{Q_0}$ (“topological charge” or “monopole charge”), which is the same as an action of the torus $T_{\mathrm{top}} = \mathrm{Hom}(\pi_1(G), \mathbb{C}^\times)$. The character ν of G is a cocharacter of T_{top} , and the *scheme-theoretic* fixed locus of this cocharacter has coordinate ring

$$\mathbb{C}[Y_{\mathfrak{f}}^\nu] = A_F / \langle (A_F)_m : m \neq 0 \rangle,$$

where $(A_F)_m$ is the subspace on which ν acts with weight m , i.e. the sum of charge spaces $A_F^{(d)}$ with $\nu \cdot d = m$. taken: nilpotents are kept.

1.3. The common coefficient ring. Both sides are algebras over

$$R = H_{G \times F}^*(\text{pt}) = \mathbb{C}[\mathfrak{t} \oplus \mathfrak{f}]^W,$$

the ring of symmetric functions in the Chern roots x_{ia} of the gauge bundles together with the flavor parameters. On the Higgs side R acts by the Kirwan map $\phi_H : R \rightarrow H_F^*(X)$ (Chern classes of tautological bundles). On the Coulomb side $R = H_{G(\mathcal{O}) \times F}^*(\text{pt})$ is the charge-zero part of A_F , so it maps to $\mathbb{C}[Y_f^\nu]$ by ϕ_C . The content of the conjecture is that the *relations* agree.

Theorem 1.1 ([1, Theorem 1.1]). *For every finite quiver, every $\mathbf{v}, \mathbf{w}$, every commuting flavor torus F and every regular character ν , both ϕ_H and ϕ_C are surjective with the same kernel. Hence there is a canonical isomorphism of graded R -algebras*

$$H_F^*(X) \cong \mathbb{C}[Y_f^\nu].$$

The grading on the left is cohomological degree, on the right the BFN homological degree. The empty case ($X = \emptyset$, both rings zero) is included.

This is the conjecture as formulated by Dumanski–Krylov [2, Conj. 2.1, eq. (2.3)], building on Hikita [4], Nakajima, and Kamnitzer–Tingley–Webster–Weekes–Yacobi. Previously known cases: the Jordan quiver (Gieseker varieties, Krylov–Shlykov [5]) and finite type A, D, E_6 , with restrictions on the framing in E_7, E_8 (Dumanski–Krylov [2]). The theorem does *not* address the quantized version, nor gauge groups other than products of GL's.

2. ARCHITECTURE OF THE PROOF

Introduce the intermediate ring

$$B_F = \mathbb{C}[Y_f^{T_{\text{top}}}] = A_F / \langle A_F^{(d)} : d \neq 0 \rangle,$$

the coordinate ring of the fixed scheme of the *full* topological torus (equivalently: kill every monopole operator of nonzero charge). There are natural maps

$$\mathbb{C}[Y_f^\nu] \twoheadrightarrow B_F, \quad R \twoheadrightarrow B_F, \quad R \twoheadrightarrow H_F^*(X).$$

The proof splits into two independent halves.

Part I (§§3–7 of the paper): $B_F \cong H_F^*(X)$, i.e. $\ker(R \rightarrow B_F) = \ker(R \rightarrow H_F^*(X))$.

Part II (§8): $\mathbb{C}[Y_f^\nu] \rightarrow B_F$ is an isomorphism, i.e. the ideal generated by nonzero- ν -weight monopoles already contains every nonzero-charge monopole.

Part I is itself a three-stage argument: prove one inclusion of ideals in general; prove equality in a special situation (loops at every vertex, determinant stability) by a size count; deform back.

3. PART I, STAGE 0: GENERATORS ON BOTH SIDES

Both B_F and $H_F^*(X)$ are quotients of R :

- *Higgs*: Kirwan surjectivity for quiver varieties (McGerty–Nevins [6]), including the equivariant version. This also gives $H^{\text{odd}}(X) = 0$.

- *Coulomb*: BFN abelianization embeds A_F into $(\text{Frac}(P_F)[X_*(T)])^W$ where $P_F = \mathbb{C}[x_{ia}] \otimes \mathbb{C}[\mathfrak{f}]$, and Weekes's generation theorem [7] says A_F is generated over R by the *dressed minuscule monopoles*: for a coweight λ all of whose entries are in $\{0, 1\}$ (or all in $\{0, -1\}$) and a W_λ -invariant polynomial p ,

$$(1) \quad M_{\lambda,p} = \sum_{\lambda' \in W\lambda} e^{\lambda'} p_{\lambda'} \frac{\prod_{\beta \in \text{wt}(N)} \beta^{(-\langle \beta, \lambda' \rangle)_+}}{\text{eu}_T(T_{t^{\lambda'}} \text{Gr}_G^\lambda)}.$$

Consequently the ideal $\langle A_F^{(d)} : d \neq 0 \rangle$ is generated by the nontrivial minuscule monopoles, and any relation $u \in \ker(R \rightarrow B_F)$ can be written as a finite sum of *neutral words*: products of minuscule monopoles of total charge zero.

4. PART I, STAGE 1: THE STABLE-PAIR TEST (ONE INCLUSION)

Proposition 4.1 ([1, Prop. 4.1]). *For all data in Theorem 1.1, $\ker(R \rightarrow B_F) \subseteq \ker(R \rightarrow H_F^*(X))$; so the coefficient map induces a graded surjection $B_F \twoheadrightarrow H_F^*(X)$.*

Idea. Take a Coulomb relation $u \in \ker(R \rightarrow B_F)$. By stage 0, u lies in the ideal of A_F generated by the nontrivial minuscule monopoles, and since u has charge zero we may discard all other charge components, so

$$u = \sum_w M_w, \quad M_w = M_{\lambda_{w,1}, p_{w,1}} \cdots M_{\lambda_{w,k_w}, p_{w,k_w}},$$

where every letter is a nontrivial (hence nonzero-charge) minuscule monopole and the charges (images in $\pi_1(G)$) of the letters in each word sum to zero. There are no separate coefficients: for $r \in R$, which is W -invariant, (1) gives $r M_{\lambda,p} = M_{\lambda, rp}$, so coefficients are absorbed into the dressings. One wants to show u restricts to zero on $D = \mu^{-1}(0)^{\nu\text{-st}}$. The paper does this by building a finite-dimensional “test” that sees both the identity lattice (where u acts by cup product) and the first step of each word (which it kills).

- (1) *Finite correspondences* (§3). Truncate $N(\mathcal{K})$ to $L = t^{-l}N(\mathcal{O})/t^lN(\mathcal{O})$ and, for a bounded family B of lattices $b = gG(\mathcal{O})$, let $S_B \rightarrow L$ be the family of truncated lattices $E_b = gN(\mathcal{O})/t^lN(\mathcal{O})$. A word in minuscule monopoles becomes a chain of incidence correspondences between such families, and the paper proves (by a localization argument that is faithful because the relevant Borel–Moore homology is free over $H_{T \times F}^*(\text{pt})$) that the relation $u = \sum_w M_w$ in A_F becomes an identity $u \cdot \rho_Z = \sum_w \Psi_w$ of morphisms of constructible sheaves on L , where ρ_Z is restriction to the identity lattice and each Ψ_w factors through the family of the *first* lattice of the word w .
- (2) *The test* (§4). On $\tilde{L} = L \times D \times P_-$ define the function

$$W(n, s, p, y) = \text{Res}(\langle p + y, n \rangle - \langle y, s \rangle), \quad (s, p) \in D,$$

and test a sheaf $\mathcal{F}$ on L by taking $G \times F$ -equivariant cohomology of the vanishing cycles $\Phi_W(\text{pr}^*\mathcal{F})$. The critical locus of W restricted to a lattice family is exactly the locus where the stable pair (s, p) extends across the lattice modification to a pair of sections of the associated bundle on $\mathbb{P}^1$.

- (3) *Stability kills the first step*. If $\mathcal{P}$ is a G -bundle on $\mathbb{P}^1$ with a section of $\mathcal{P} \times_G T^*N$ whose value at ∞ is ν -stable with trivial stabilizer, then $\deg \mathcal{L}_\nu \geq 0$, with equality only if $\mathcal{P}$ is trivial and the section is constant. (Proof: a ν^m -semi-invariant nonzero at ∞ gives a nonzero section of $\mathcal{L}_\nu^m$; in degree zero the induced map $\mathbb{P}^1 \rightarrow T^*N//_\nu G$ is constant because the GIT polarization is relatively ample over the affine quotient.) Since each word is neutral, the ν -degrees of its steps sum to zero, so (by commutativity) one can put a step of nonpositive ν -degree and nonzero charge first. Its critical locus is empty, so its test vanishes.

- (4) *The identity lattice survives.* On the neutral family the test is computed in a quadratic normal form: after a change of variables the function becomes $\langle y', a' \rangle$ (this uses $\mu(s, p) = 0$ to show the critical value is zero), so vanishing cycles are a rank-one local system shifted by $2r$, and the test of ρ_Z becomes ordinary restriction $H_{G \times F}^*(U \times D) \rightarrow H_{G \times F}^*(D)$, which is split surjective.

Combining: $u|_D \smile j^* = 0$ with j^* split, so $u|_D = 0$ in $H_{G \times F}^*(D) = H_F^*(X)$.

5. PART I, STAGE 2: ADD LOOPS AND COUNT

Now suppose temporarily that Q has a loop at every vertex with $v_i > 0$, there is no flavor, and $\nu = \nu_{\det} = \prod_i \det_{V_i}$ (which is always regular). Write B_0 for the no-flavor version of B_F .

5.1. The Laurent model. The key observation is that one loop at a vertex contributes to the numerator of (1) the factor $\prod_{a \in I, b \notin I} (x_{ib} - x_{ia})$, which is exactly the Euler class of the tangent space of the Grassmannian Gr^λ at the fixed point $t^{\lambda'}$. So with a loop at every vertex, the denominators in (1) cancel and A_0 embeds in a *polynomial* Laurent ring:

Proposition 5.1 ([1, Prop. 5.1]). *Let $\mathcal{B}'$ be the multiset of weights of N with one loop per vertex removed, and set $r_\lambda = e^\lambda \prod_{\beta \in \mathcal{B}'} \beta^{(-\langle \beta, \lambda \rangle)_+}$. Then*

$$A_0 = \left(\bigoplus_{\lambda \in \mathbb{Z}^\Omega} P r_\lambda \right)^W, \quad P = \mathbb{C}[x_{ia}].$$

Both inclusions need proof; the nontrivial one ($\supseteq$) is an induction on $\|\lambda\|^2$ using a rearrangement-inequality argument to show that each r_λ with arbitrary stabilizer-invariant dressing is a product of two shorter ones plus lower terms.

5.2. Filtration by multipartitions and cut ideals. A multipartition $m = (m_{i,s})$ of $\mathbf{v}$ (i.e. $\sum_s s m_{i,s} = v_i$) determines a charge-zero coweight λ_m with entries $s - 1$ on $m_{i,s}$ slots of color i . The paper filters B_0 by the number $|l|$ of nonzero parts and shows that the associated graded piece for m is a quotient of

$$(P_m/J_m)^{W_m}, \quad P_m = \mathbb{C}[x_d : d \text{ a nonzero part}],$$

where J_m is the *cut ideal* generated by

$$q_S = \prod_{d \in S} x_d^{w_{i(d)}} \prod_{d \in S, e \notin S} (x_d - x_e)^{b_{de}}, \quad \emptyset \neq S \subseteq \{\text{parts}\},$$

with $b_{de} = c_{i(d)i(e)} \min(s(d), s(e))$ and c_{ij} the number of arrows between i, j in either direction (after removing the chosen loops). The relations q_S come from the vanishing in B_0 of products $M_\alpha M_\beta$ with β of nonzero positive charge: the excess weight factor in $r_\alpha r_\beta / r_{\lambda_m}$ is precisely q_S . These cut ideals are the cocircuit ideals familiar from toric hyperkähler cohomology (Hausel–Sturmfels). The upshot is a coefficientwise bound

$$(2) \quad \text{Hilb}(B_0; q) \leq \sum_m q^{\delta_m} \text{Hilb}((P_m/J_m)^{W_m}; q),$$

with explicit shifts δ_m .

5.3. Matching Hausel's formula. The paper computes $\text{Hilb}((P_m/J_m)^{W_m})$ by realizing P_m/J_m as the T_m -equivariant cohomology of an open "all slots reachable from the framing" locus in a representation space, stratified by the set of unreachable slots. This gives a recurrence whose generating function is a ratio of two q -series, and that ratio is exactly the right-hand side of Hausel's arithmetic-Fourier-transform formula [3] for the Poincaré polynomial of $\mu^{-1}(\text{id})//G$. A hyperkähler rotation identifies that affine quotient with $X_{\det}$ topologically. Hence the right side of (2) equals $P_{X_{\det}}(q)$. Together with the surjection of stage 1:

Proposition 5.2 ([1, Cor. 6.6]). *With a loop at every vertex and $\nu = \nu_{\det}$, the coefficient map is an isomorphism $B_0 \cong H^*(X_{\det})$.*

6. PART I, STAGE 3: REMOVE THE LOOPS, THEN VARY EVERYTHING

6.1. Loop removal by a mass deformation. For an arbitrary quiver Q , let Q^+ add one loop at each vertex, and let an auxiliary $H = \mathbb{C}^\times$ act with weight 1 on the new loops only, with equivariant parameter z . The previous stage plus a flavor-lift lemma (see below) gives $B_H^+ \cong H_H^*(X^+)$ as $R_0[z]$ -algebras. Now localize at the maximal ideal of $(x, z) = (0, 1)$, i.e. near zero gauge parameters and mass one.

- *Coulomb side:* the extra loop contributes to each minuscule monopole the factor $E_\lambda = \prod (z + x_{ib} - x_{ia})^{(\lambda_{ia} - \lambda_{ib})_+}$, which equals 1 at $(0, 1)$, hence is a unit after localization. So the localized Coulomb algebras of Q and Q^+ agree, compatibly with the charge grading.
- *Higgs side:* Brion's localization theorem on the stable prequotient D^+ (not on the quotient, where H -fixed points need compensating gauge transformations) shows restriction to $(D^+)^H = D$ is an isomorphism after inverting characters nontrivial on H . Stability restricts correctly under the closed embedding $T^*N \subset T^*N^+$.

This gives equality of the two ideals in R_0 after localizing at the graded origin. Both ideals are homogeneous, and a homogeneous ideal is determined by its localization at the origin (if $sh \in I$ with $s(0) \neq 0$, take the component of degree $\deg h$). So the ideals are equal globally. This is Prop. 7.3: the point is that one does not need to know that a "generic" isomorphism specializes.

6.2. Other stabilities. All regular ν give quotients with the same Betti numbers (Lemma 7.1): a hyperkähler rotation identifies X_ν with the affine quotient $\mu^{-1}(c\nu)//G$, and the compactly supported Betti numbers of the fibers of $\{\mu = \zeta\}//G \rightarrow \mathfrak{c}$ are constant on a dense open set that every real ball meets. So for any regular ν , the surjection $B_0 \twoheadrightarrow H^*(X_\nu)$ of stage 1 is between spaces of the same finite dimension.

6.3. Flavor lift. If $B_0 \cong H^*(X)$ then $B_F \cong H_F^*(X)$ for any F : $H_F^*(X)$ is free over $\mathbb{C}[\mathfrak{f}]$ (Leray–Hirsch, using $H^{\text{odd}} = 0$), B_F base-changes to B_0 (an even-strata argument on the space of triples), and a graded Nakayama argument kills the kernel.

This completes Part I: $B_F \cong H_F^*(X)$ for all data.

7. PART II: COCHARACTER-FIXED VERSUS TORUS-FIXED

Proposition 7.1 ([1, Prop. 8.4]). *For regular ν , the quotient map $\mathbb{C}[Y_{\mathfrak{f}}^\nu] \rightarrow B_F$ is an isomorphism.*

Equivalently: every nontrivial minuscule monopole already lies in the ideal generated by monopoles of nonzero ν -weight. If $\nu \cdot \mathbf{b} \neq 0$ for all $0 < \mathbf{b} \leq \mathbf{v}$ this is immediate; the content is the hyperplane case.

Idea: complementary monopoles. Adjoin a framing vertex ∞ of dimension 1 with w_i arrows $i \rightarrow \infty$, making an *unframed* quiver $\widehat{Q}$ with dimension vector $\mathbf{d} = (\mathbf{v}, 1)$ and the same representation space. The extra $\mathbb{C}^\times$ is central and acts trivially, so

$$A_{\widehat{Q}} \cong A_F \otimes \mathbb{C}[\xi, z^{\pm 1}],$$

where z is the invertible monopole of charge $\mathbf{d}$ (“translate every coweight entry by one”). Extend ν by $\widehat{\nu}_\infty = -\sum \nu_i v_i$ so that $\widehat{\nu} \cdot \mathbf{d} = 0$; then z has weight zero and the cocharacter quotients are compatible. Translation by z gives the identity

$$M_{\mathbf{b},p}^+ = z M_{\mathbf{d}-\mathbf{b},p}^-$$

between a positive minuscule monopole of charge $\mathbf{b} = (\mathbf{b}', 0)$ and the *complementary* negative one. So if $M_{\mathbf{b},p}^+$ survived in the $\widehat{\nu}$ -fixed quotient, so would $M_{\mathbf{d}-\mathbf{b},p}^-$.

Now use the polynomial-form cohomological Hall algebra of the tripled quiver (Jindal–Neguț), together with Tsymbaliuk’s difference-operator realization of the shuffle algebra, to prove (Lemma 8.2): every minuscule monopole of charge $\mathbf{b}$ is an R -combination of words whose factors have charges $\gamma_h \in \Delta_{\widehat{Q}}$ (the support of Kac’s polynomials $A_{\widehat{Q},\gamma}(q)$, i.e. dimension vectors admitting absolutely indecomposable representations) summing to $\mathbf{b}$. A word survives the cocharacter quotient only if each factor has $\widehat{\nu}$ -weight zero. So survival of both complementary monopoles decomposes $\mathbf{d} = \sum_h \gamma_h$ into at least two Kac-support vectors each orthogonal to $\widehat{\nu}$.

Finally (Lemma 8.3, essentially Crawley-Boevey’s lifting lemma): for $\gamma \in \Delta_{\widehat{Q}}$ with $\widehat{\nu} \cdot \gamma = 0$, the hyperkähler moment fiber $\mu_{\mathbb{C}}^{-1}(0) \cap \mu_{\mathbb{R}}^{-1}(\tau_{\widehat{\nu}})$ is nonempty. The direct sum of such points for the blocks γ_h is $\widehat{\nu}$ -semistable with a stabilizer containing $k \geq 2$ independent scalar circles; after removing the ineffective diagonal one, the stabilizer is still positive-dimensional. This contradicts regularity of ν (including when the stable locus is empty: it would produce a semistable point where there is none). Hence no nontrivial minuscule monopole survives, and the two ideals coincide.

8. ASSESSMENT

Inputs from the literature. BFN abelianization and minuscule generation (BFN, Weekes); Kirwan surjectivity for quiver varieties (McGerty–Nevins); Hausel’s Betti-number formula; hyperkähler rotation and the topological Kempf–Ness theorem; Brion’s subgroup localization; constructible-sheaf formalism for vanishing cycles (Maxim–Schürmann, Massey); Kac polynomials and Crawley-Boevey’s lifting criterion; the polynomial-form CoHA and shuffle-to-difference-operator maps (Jindal–Neguț, Tsymbaliuk). The paper cites Krylov–Shlykov [5, Remark 1.15] for related deformation arguments and Dumanski–Krylov [2, Prop. 2.3] for the prequotient localization viewpoint.

What appears to be new.

- (1) The *stable-pair vanishing-cycle test* (§§3–4): a finite-dimensional device turning a Coulomb relation into a statement about sections of bundles on $\mathbb{P}^1$, where stability of the value at ∞ forces the degree inequality that kills the first step of every neutral word.
- (2) The *added-loop Laurent model* (§5): one loop per vertex cancels the Grassmannian Euler denominators, giving a polynomial model in which an explicit multipartition filtration and cut ideals bound B_0 .
- (3) The *reachability stratification* (§6) that computes the cut quotients and matches Hausel’s q -series exactly.
- (4) The *homogeneity descent* (§7) that removes the auxiliary loop without any genericity assumption.

- (5) The *complementary-monopole argument* (§8) for the cocharacter-versus-torus comparison, using Kac support plus regularity.

Verification status. After these notes were drafted, each of §§2–8 was read independently by a separate Claude instance acting as an adversarial referee, recomputing every formula it could and checking citations against the sources where they could be fetched (BFN18, Weekes, Hoskins, McGerty–Nevins, Tsymbaliuk, Jindal–Neguț). *No mathematical error and no load-bearing gap was found.* A dozen explicit computations (Jordan $v = 2$, $w = 1, 2$; A_1 and A_2 with loops; the GL_2 loop cancellation; the Tsymbaliuk kernel and sign table; the central splitting in §8) all agree. The issues found are presentational: one convention that should be stated (the grading is BFN’s $N_{\mathcal{O}}$ -relative one, not the symmetric $2\Delta(\lambda)$; they differ by a function of charge), two steps in §4 that deserve a paragraph each (equivariant Thom–Sebastiani; compatibility of two vanishing-cycle identifications, which in any case only matters up to a unit), and a handful of citations to make precise. The full list is in `VERIFICATION.md` in the repository. This is a careful machine reading, not a human referee report; it raises the prior that the paper is correct without settling it.

Scope. The theorem is for $G = \prod GL(V_i)$ and N of quiver type. Loops, multiple arrows, arbitrary flavor, and all regular stabilities are allowed. Not covered: the quantized statement; other reductive gauge groups (the paper cites Hoang–Krylov–Matvieievskiy, Ex. 6.7, for a failure outside the quiver setting); K -theory.

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